GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes
Please check MSDS and review the ISEF rules for Hazardous Chemicals, Activities, or Devices to determine the safety requirements for working with chemicals.						
ISEF rewview rules for Hazardous Chemicals						
	Use "control F "to search					
			Acetic Acid	1.00	1 -3	Vinegar - see separate rating
1- Severe	Lab/RRI		Acetone	2.00	2-3	
2-Serious	Home*/School/RRI		Agar		no GHS hazards	
3-Moderate	Home*/School/RRI		Agarose (powder)		no GHS hazards	
4-Slight	Home*/School/RRI		Agarose (solution)		no GHS hazards	
5-Minimal	Home*/School/RRI		Ammonia (household)	2.00	2	
none			Ammonium Chloride	2.00	2 & 4	
			Amylase		no GHS hazards	
			Atrazine	1.00	1 & 2	
*	Substance has specific disposal instructions- see MSDS- Section 13		Benedict's Reagent (Solution)	2.00	2 & 3	
			Benzoic acid	1.00	1-3	
			Biuret (Reagent)	2.00	2 & 3	
**			Bromothymol blue*		no GHS hazards	do not empty into drains
			Bismuth	1.00	flammable solid	
	All projects using chemicals NOT for their intended purpose and with GHS safety ratings between 2--5 in any of the classifications, must be conducted in a school or RRI laboratory setting. A chemical is NOT used for its intended purposes when it is mixed with other chemicals, changes temperature, larger volumes are used, etc. - ISEF Rulebook, pg. 18		Cadmium chloride	1.00	1-3	
			Calcium Chloride	2.00	2	
			Calcium hydroxide (lime water)	1.00	1-3	
			Calcium- metal	2.00	2	flammable in contact with water
			Chlorine	1.00	1 & 2	
			Citric acid	2.00	2-3	
			Cobalt (II) chloride	1.00	1-2	
			Copper		no GHS hazards	
			Copper (II) chloride	1.00	1-2	
			Copper (II) sulfate pentahydrate	1.00	1 & 4	
			Copper nitrate hydrate	1.00	1-2	
			Crystal Violet	1.00	1,2,4	
			Cyclohexane	1.00	1-4	
GHS ratings were obtained from Fisher, Sigma/Aldrich and AK Scientific MSDS sheets			Ethyl Alcohol < 10%		no GHS Hazard	
			Ethyl Alcohol 10-99%	2.00	2 & 3	Health Hazard & Highly Flammable
Safety Data Sheets - use this link to search for additional chemicals or additional requirements			Filings, zinc, magnesium, iron	2.00		Health Hazard & Highly Flammable
			Gelatin		no GHS hazards	
			Glyphosate	1.00	1 & 2	
			Graphite		no GHS hazards	
			Guaiacol (0.2% solution)	1.00	1-2	solution contains 70% isopropyl alcohol
			Hydrochloric Acid	1.00	1,3	
			Hydrogen peroxide- 10% -20%	2.00	2-4	food grade
			Hydrogen peroxide- 3-5%		no GHS Hazard	
			Hydrogen peroxide- 30% or higher	1.00	1,2,4	
			Iodine solutions (0.02M)	1.00	1-4	
			Iron chloride sodium citrate complex	1.00	1,2,4	
			Iron Nitrate, 0.1 M	1.00	1-2	
			Isopropyl Alcohol(isopropanol, 2- propanol) 70-99%	2.00	2-3	
			Lauric Acid (Dodecanoic acid)	1.00	1-2	
			Lead	1.00	1-2	
			Lead nitrates	1.00	1-4	
			lemon juice (5-6% citric acid)		no GHS hazards	
			Litmus Solution / Paper		no GHS hazards	
			Lugol's iodine [Iodine (aq) (tincture)]	2.00	2	
			Luminol	3.00	3	
			Magnesium oxide*			must be disposed of following local regulations
			Magnesium sulfate*			must be disposed of following local regulations
			Methanol (methyl alcohol)	1.00	1-3	
			Methyl orange	2.00	2-3	
			Methylene Blue*	4.00	4	must be disposed of following local regulations
			Nickel	1.00	1-2	
			Nitrate	1.00	1-3	
			Nitric Acid	1.00	1 & 3	
			Phenolphthalein	2.00	2	
			Phosphoric Acid	1.00	1 & 3	
			Potassium hydroxide	1.00	1,3,4	
			Potassium Iodate	1.00	1,2,4	
			potassium Iodide	1.00	1	
			Potassium nitrate	3.00	3	
			Potassium permangaate	1.00	1-4	
			Potassium thiocyanate*	3.00	3 & 4	must be disposed of following local regulations
			Silver		no GHS hazards	
			Silver nitrate	1.00	1 & 2	
			Sodium acetate		no GHS hazards	
			Sodium alginate	2.00	2 & 3	
			Sodium bicarbonate (baking soda)		no GHS hazards	specific for baking soda
			Sodium chloride - table salt		no GHS hazards	
			Sodium hydroxide (i.e. caustic soda)	1.00	1 &3	
			Sodium hypochlorite (Bleach)	2.00	2	this GHS is specific for Bleach
			Sodium polyacrylate (diaper compound)	2.00	2	
			Sodium thiosulfate*		no GHS hazards	must be disposed of following local regulations
			Starch		no GHS hazards	
			Sulfur	2.00	2	
			Sulfuric Acid	1.00	1	
			Tin		no GHS hazards	
			Titanium dioxide	2.00	2	
			Trisodium phosphate	2.00	2 &3	
			Universal Indicator	2.00	2	
			Vinegar (20-25% Acetic Acid)		no GHS hazards	
			Zinc nitrate	2.00	2,3,4	

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes

GHS-Hazard -KEY	color- KEY		Chemical	Lowest GHS Rating	GHS ratings all categories	Notes